



International University, VNU-HCMC

School of Computer Science and Engineering

Lecture 8 Ethical and Legal Considerations in Big Data Analytics

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Purpose of the lecture

- Highlight key ethical and legal challenges in Big Data Analytics
- Introduce legal frameworks and regulatory constraints on data usage
- Understand issues of privacy, security, and data protection
- Recognize bias, fairness, and transparency in analytics models
- Promote responsible and ethical use of big data in practice

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Warm-up question

- Who should be responsible when a data analytics system causes harm: the developer, the company, or the algorithm?



- Is removing personal identifiers enough to guarantee data privacy? Why or why not?



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Outlines



Motivation and Importance of Ethics in Big Data Analytics



Data Privacy and Personal Data Protection



Data Security and Access Control



Bias, Fairness, and Transparency in Analytics Models



Legal and Regulatory Frameworks in Big Data

Responsible Use of Big Data Analytics

Case Studies and Real-World Incidents

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Motivation and Importance of Ethics in BDA

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Why Ethics Matters in Big Data



The Power and Risks of Big Data



Real-World Impact of Data Decisions



From Technical Problem to Ethical Problem



Why Ethics is Critical in Big Data

Key Ethical Questions in Big Data

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Why Ethics Matters in Big Data?

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Big Data enables organizations to collect, store, and analyze massive amounts of personal and behavioral data



Decisions are increasingly data-driven and automated (AI/ML systems)



These decisions can directly impact individuals and society

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Why Ethics Matters in Big Data?

- Ethics ensures that data is used in a way that is:



Fair



Responsible



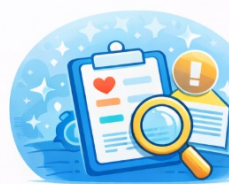
Respectful
of human rights

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The Power and Risks of Big Data

Power



Improve decision-
making (e.g., healthcare
diagnosis, fraud detection)



Enable personalization
(e.g., recommendation
systems)



Support innovation
and economic
growth

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The Power and Risks of Big Data

Risks



Misuse of
personal data



Unfair or
biased decisions



Lack of
transparency
("black-box" models)



Loss of trust
from users



More data = more power → requires more responsibility



Real-World Impact of Data Decisions

- Big Data systems influence:



Loan approval



Hiring
decisions



Insurance
pricing



Medical treatment
recommendations



Real-World Impact of Data Decisions

- A wrong or biased model can:



Discriminate
against certain groups



Make incorrect or
harmful decisions

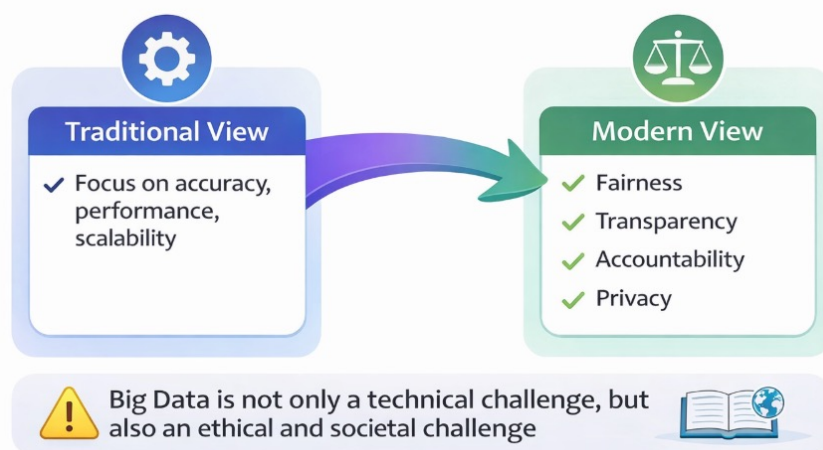


Affect people's
lives at scale

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From Technical Problem to Ethical Problem



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Why is Ethics Critical in Big Data?



Protect individual rights and privacy



Ensure fair and unbiased outcomes



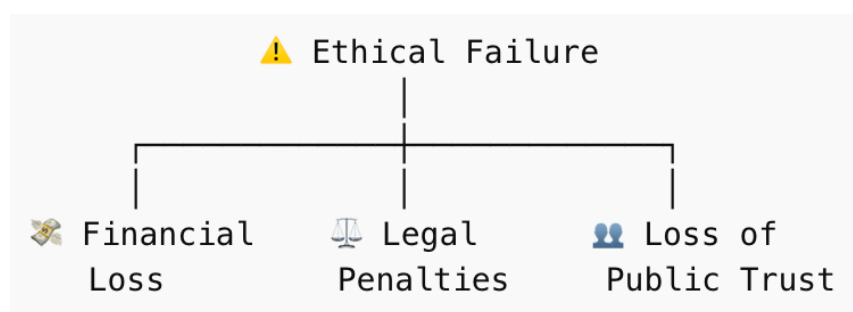
Build trust in data-driven systems



Prevent legal and reputational risks for organizations



Why is Ethics Critical in Big Data?



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Key Ethical Questions in Big Data

-  Is it acceptable to collect and use this data?
-  Who owns the data?
-  Is the model fair to all users?
-  Can decisions be explained?
-  Who is responsible when something goes wrong?

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Data Privacy and Personal Data Protection

-  What is Data Privacy?
-  What is Personal Data?
-  Types of Privacy Risks in Big Data
-  Data Lifecycle and Privacy Risks
-  Key Privacy Principles
-  Privacy-Preserving Techniques
-  Challenges in Big Data Privacy
-  Real-World Example

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What is Data Privacy?

- Data privacy refers to the right of individuals to control their personal information
- It concerns:

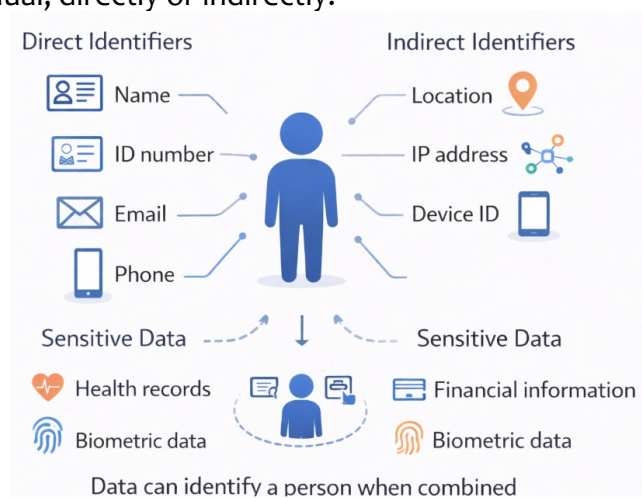


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What is Personal Data?

- Personal data is any information that can identify an individual, directly or indirectly:



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Types of Privacy Risks in Big Data



- Unauthorized data collection (without consent)



- Data leakage or exposure



- Re-identification of anonymized data

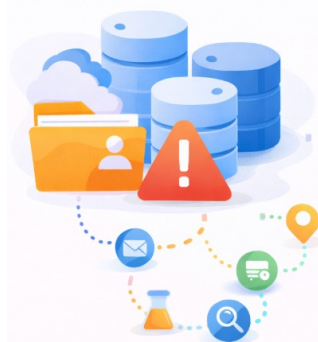


- Data misuse beyond the original purpose



Types of Privacy Risks in Big Data

- Big Data increases risk due to:



- Large volume
- Data integration from multiple sources



Data Lifecycle and Privacy Risks

- Data goes through multiple stages
- Privacy risks can occur at any stage



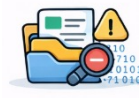
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Key Privacy Principles



- **Consent:** users must agree to data collection



- **Purpose limitation:** data used only for specific purposes



- **Data minimization:** Collect only necessary data



- **Transparency:** users know how data is used



- **Accountability:** organizations are responsible for data handling



- These principles guide ethical data usage

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Privacy-Preserving Techniques

Anonymization



- Remove personal identifiers
- Risk: may still be re-identified

Pseudonymization



- Replace identifiers with artificial IDs

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Privacy-Preserving Techniques



Encryption

- Protect data during storage and transmission

+



Differential Privacy

- Add noise to the data to prevent identifying individuals
- No technique is perfect
→ often need a combination

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Challenges in Big Data Privacy



- ✓ Difficulty in fully anonymizing large datasets



- ✓ Data from multiple sources can be linked together

- ✓ More privacy → less data utility
- ✓ More accuracy → higher privacy risk



- ✓ Balancing privacy vs usefulness is a key challenge

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Real-World Example



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Data Security and Access Control

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-  What is Data Security?
-  Why Data Security is Critical in Big Data?
-  Common Security Threats
-  Data Security Across the Data Lifecycle
-  Key Security Techniques
-  Access Control Models
-  Monitoring and Auditing
-  Challenges in Big Data Security
-  Real-World Example

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What is Data Security?

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Protect data from unauthorized access, breaches, and attacks



Confidentiality
No unauthorized access

Integrity
Data is accurate and not altered

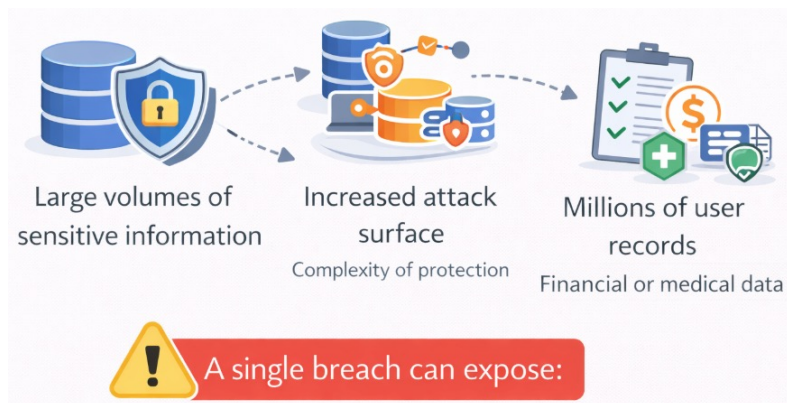
Availability
Data is accessible when needed

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Why Data Security is Critical in Big Data?

- Big Data systems store



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Common Security Threats

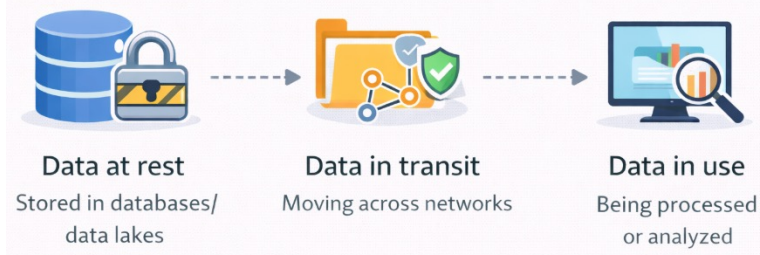


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Data Security Across the Data Lifecycle

Security applied at all stages:



- Each stage requires different protection mechanisms

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Key Security Techniques

Encryption

Protect data:



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Key Security Techniques

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Authentication

Verify the identity of users/systems



Passwords

(e.g, database encryption)



Multi-factor authentication (MFA)

(e.g, HTTPS, SSL/TLS)

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Key Security Techniques

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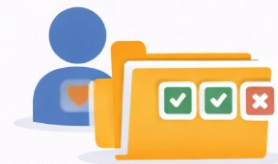
Authorization

Control what users are allowed to do



Authorization

= Who are you?



Authorization

= What can you do?

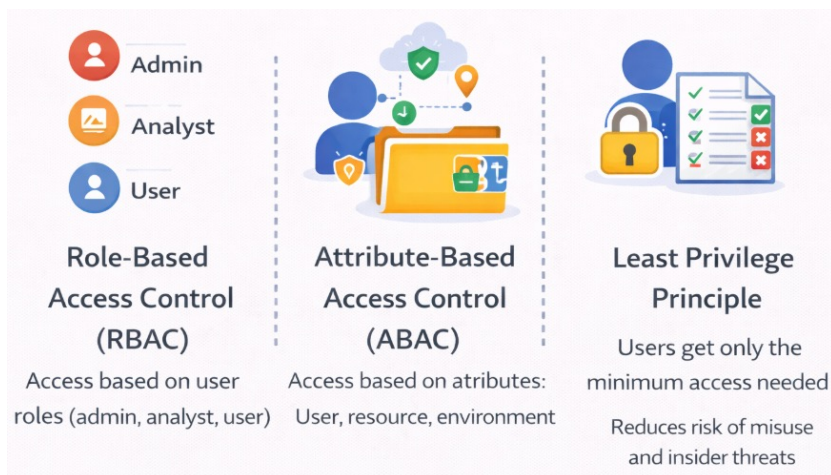


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Access Control Models

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Monitoring and Auditing

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- Track who accesses data and what actions are performed
- Maintain logs and audit trails



Helps to:



Detect suspicious behavior



Investigate incidents



Ensure accountability

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Challenges in Big Data Security

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Distributed systems (cloud, clusters) are harder to secure



Managing access across many users and systems



Real-time data processing increases risk exposure



Balancing: Security vs performance

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Real-World Example

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- A company stores user data without encryption
- Hackers gain access → massive data breach



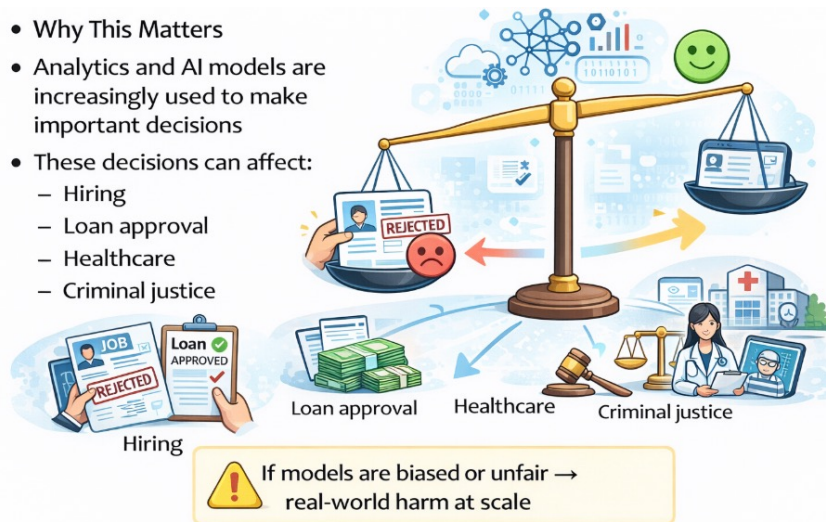
Lesson: Lack of basic security measures can lead to serious consequences

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Bias, Fairness, and Transparency in Analytics Models

- Why This Matters
- Analytics and AI models are increasingly used to make important decisions
- These decisions can affect:
 - Hiring
 - Loan approval
 - Healthcare
 - Criminal justice



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Bias, Fairness, and Transparency in Analytics Models

- ? Why This Matters?
- What is Bias in Data and Models?
- Types of Bias
- What is Fairness?
- Fairness Challenges
- What is Transparency?
- Black-Box vs Explainable Models
- Techniques to Improve Fairness and Transparency

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Why This Matters?

Analytics and AI models are increasingly used to make important decisions



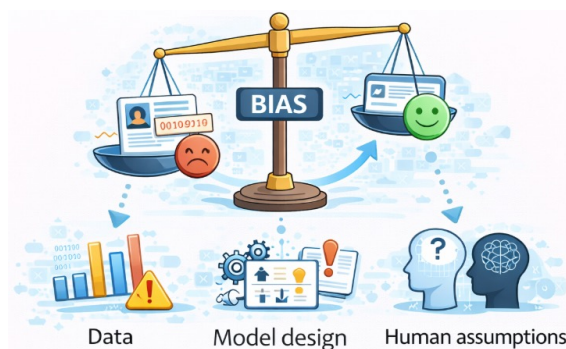
If models are biased or unfair →
real-world harm at scale

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What is Bias in Data and Models?

- Bias refers to systematic errors that lead to unfair outcomes
- It can originate from:



Models learn from data → biased
data leads to biased models

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Types of Bias



Data Bias

- Training data is not representative
- Example: the dataset mostly contains one group



Sampling Bias

- Certain populations are underrepresented

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Types of Bias

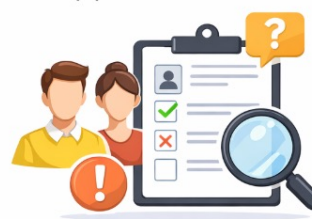
Algorithmic Bias

- Model favors certain patterns or groups



Human Bias

- Bias introduced during feature selection or labeling
- Bias can occur at any stage of the pipeline



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What is Fairness?



- Fairness means treating individuals or groups equitably
- A fair model should:
 - Not to discriminate based on sensitive attributes
 - Provide equal opportunities across groups



• Example:

- ✓ Same qualifications → same chance of approval



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Fairness Challenges

- Different definitions of fairness:



Equal outcomes

Equal opportunities



- Trade-offs:
 - Fairness vs accuracy



- Some attributes are:
 - Sensitive (e.g., gender, race)
 - But still indirectly influences results



No single “perfect” fairness solution

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What is Transparency?



Transparency refers to how understandable and explainable a model is.

Two levels:

- **Model transparency:**
how the model works



- **Decision transparency:**
why a specific decision was made



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What is Transparency?

- Important for:



Trust



Accountability



Debugging models

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Black-Box vs Explainable Models



Black-box models

- High accuracy
- Low interpretability



Explainable models

- Easier to understand
- More transparent decisions

Accuracy  Interpretability

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Techniques to Improve Fairness and Transparency



- Data preprocessing:
 - Balance datasets



- Explainability tools:
 - Feature importance
 - Model explanation methods (XAI)



- Model evaluation:
 - Check performance across groups

Goal: Make models more  trustworthy and accountable

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Real-World Example

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A hiring model trained on historical data



Past hiring favored one group



The model continues to prefer that group



Lesson: Models can amplify existing biases

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Legal and Regulatory Frameworks in BD

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Why Legal Frameworks are Important?



What Do Data Laws Regulate?



Key Principles in Data Protection Laws



User Rights Under Data Protection Laws



Examples of Regulations



Compliance Requirements for Organizations



Consequences of Non-Compliance



Challenges in Big Data Regulation

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Why Legal Frameworks are Important?



- Big Data involves collecting and processing large amounts of personal data



- Without regulations, organizations may:
 - Misuse data
 - Violate user rights
- Legal frameworks ensure:
 - Protection of individuals
 - Accountability of organizations
 - Standardized rules for data usage

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What Do Data Laws Regulate?

- Legal frameworks define:



- What data can be collected
 - How data can be used and processed
 - How long can data be stored
 - Who can access or share data
 - What rights individuals have
- Organizations must comply with these rules

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Key Principles in Data Protection Laws

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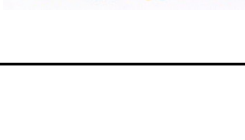
- Lawfulness, fairness, transparency



- Purpose limitation



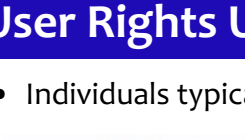
- Data minimization



- Accuracy of data



- Accuracy of data



- Storage limitation



- Accountability

These principles guide legal compliance

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User Rights Under Data Protection Laws

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- Individuals typically have the right to:



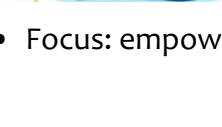
- Access their data



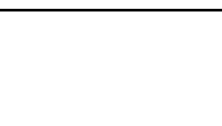
- Correct inaccurate data



- Delete their data (“right to be forgotten”)



- Restrict or object to data processing



- Data portability (transfer data to another service)

- Focus: empowering users

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Examples of Regulations



- **GDPR (EU):**
 - Strict data protection rules
 - Strong user rights



- **CCPA (California, USA):**
 - Focus on consumer data rights
- Organizations must follow local and international regulations



- **Other national laws:**
 - Vary by country



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Compliance Requirements for Organizations



Obtain user consent



Clearly state privacy policies



Implement data protection measures



Report data breaches when required



Maintain data processing records



Compliance is not optional → it is mandatory

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Consequences of Non-Compliance

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- Legal penalties and fines
- Loss of business reputation
- Suspension of services



- Loss of customer trust

Example: Companies fined millions for violating data laws



Example: Companies fined millions for violating data laws

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Challenges in Big Data Regulation

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- Different laws across countries
- Rapid growth of technology (laws may lag behind)



- Difficulty in regulating AI decisions
- Cross-border data transfer issues



- Global data → complex legal environment

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Real-World Example

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- A company collects user data without proper consent
- Authorities investigate → heavy fine
- Lesson: Ignoring regulations can lead to serious legal consequences

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Responsible Use of Big Data Analytics

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What is Responsible Data Use?



Beyond Legal Compliance



Core Principles of Responsible Analytics



Ethical Decision-Making in Practice



Responsible AI and Analytics Lifecycle



Organizational Responsibility



Role of Data Professionals



Balancing Innovation and Responsibility

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What is Responsible Data Use?

- Responsible use means applying data analytics in a way that is:



- ✓ Ethical
- ✓ Fair
- ✓ Transparent
- ✓ Accountable

- Not only “can we do it?”
- But also “should we do it?”

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Beyond Legal Compliance



- Legal compliance = minimum requirement
- Responsible use = higher standard

Example:

- Legal: allowed to collect data
- Responsible: Collect only what is truly necessary

Key idea:

- *Being legal does not always mean being ethical*

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Core Principles of Responsible Analytics

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Ethical Decision-Making in Practice

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Before using data or deploying a model, ask:

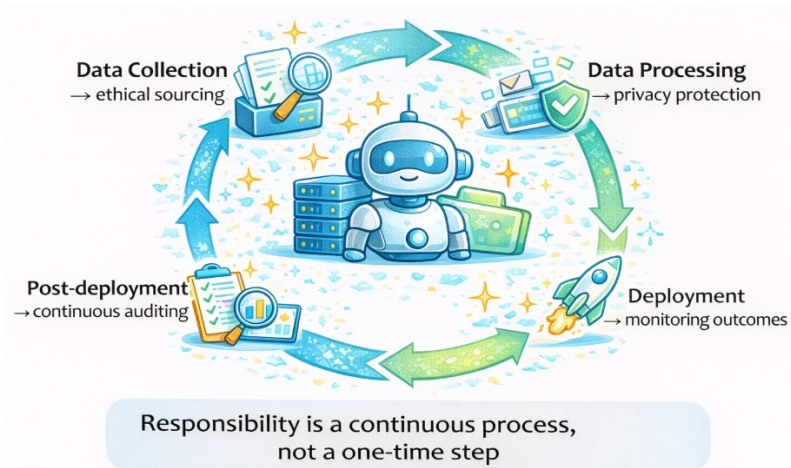
-  Does this harm any group?
-  Is the data collected with proper consent?
-  Is the outcome fair for all users?
-  Can we explain the decision?
-  Who is responsible if something goes wrong?
- Encourage critical thinking, not blind automation

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Responsible AI and Analytics Lifecycle

- Responsibility must be applied at all stages:



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Organizational Responsibility

Organizations should:



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Role of Data Professionals

- Data scientists, engineers, and analysts must:



- Understand the ethical implications of their work
- Avoid misuse of data
- Communicate the limitations of models
- Challenge questionable decisions

Responsibility is not only organizational → also individual

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Balancing Innovation and Responsibility

- Big Data drives:



- Innovation
- Business value

- Privacy
- Fairness
- Ethics

Challenge: Move fast without causing harm

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Real-World Example

- A company uses customer data to improve services
- But also ensures:



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Case Studies and Real-World Incidents

- Why Case Studies Matter
- Case 1: Data Privacy Violation
- Case 2: Data Breach Incident
- Case 3: Algorithmic Bias
- Case 4: Lack of Transparency
- Case 5: Misuse of Data
- Cross-Case Analysis

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Why Case Studies Matter

- Ethical issues actually occur
- Decisions impact real people



- Help students:
 - Connect theory with practice
 - Understand the consequences of poor decisions



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Case 1: Data Privacy Violation

- **Scenario:**
 - A company collects user data without clear consent
 - Data is shared with third parties
- **Issues:**
 - Lack of transparency
 - Violation of user privacy
- **Impact:**
 - Loss of user trust
 - Legal penalties
- Lesson: Privacy must be clear, informed, and respected

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Case 2: Data Breach Incident

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- **Scenario:**
 - Sensitive user data is stored without proper security
 - Hackers access and leak the data
- **Issues:**
 - Weak security measures
 - Poor access control
- **Impact:**
 - Financial loss
 - Reputation damage
- **Lesson:** Security is critical to protect data at all times

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Case 3: Algorithmic Bias

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- **Scenario:**
 - A hiring system trained on historical data
 - Past data favors certain groups
- **Issues:**
 - Biased training data
 - Unfair model decisions
- **Impact:**
 - Discrimination
 - Reduced fairness in hiring
- **Lesson:** Models can amplify bias if not carefully designed

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Case 4: Lack of Transparency

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- **Scenario:**
 - A loan approval system rejects applications
 - No explanation is provided
- **Issues:**
 - Black-box model
 - No explainability
- **Impact:**
 - Users cannot understand decisions
 - Reduced trust
- Lesson: Transparency is key to accountability

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Case 5: Misuse of Data

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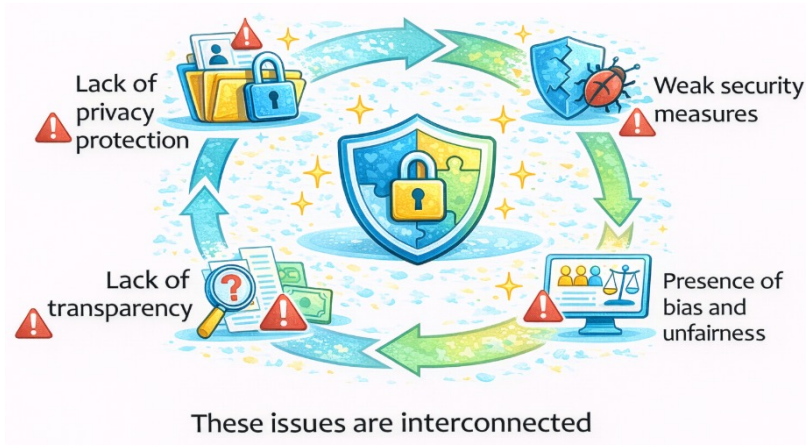
- **Scenario:**
 - Data collected for one purpose
 - Used for another purpose without consent
- **Issues:**
 - Violation of purpose limitation
 - Ethical misuse of data
- **Impact:**
 - Legal and ethical concerns
- Lesson: Data must be used only for intended purposes

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Cross-Case Analysis

- Across all cases, common problems include:

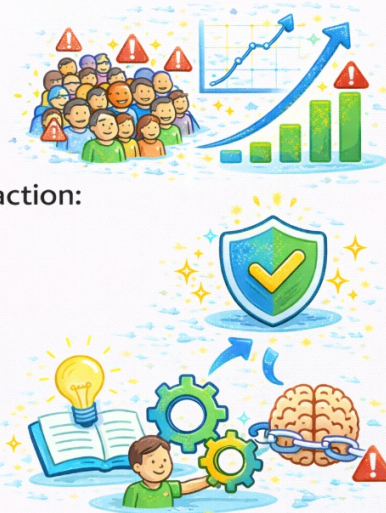


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Key Lessons from Real-World Incidents

- Ethical issues can:
 - Scale quickly
 - Affects many people
- Prevention is better than reaction:
 - Design systems responsibly from the beginning
- Ethics should be built into the system, not added later



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Case 3: Algorithmic Bias

Detecting and Mitigating Algorithmic Bias in Hiring Systems

- Context:

A company uses an AI-based hiring system trained on historical data. However, the system may inherit biases from past decisions, leading to unfair outcomes.

Tasks

- Problem Understanding
 - Explain what algorithmic bias is.
 - Describe how bias can arise from historical data in hiring.
- Dataset Exploration
 - Use a sample dataset (e.g., hiring or HR dataset) or simulate one.
 - Identify potential sensitive attributes (e.g., gender, age).

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Case 3: Algorithmic Bias

Detecting and Mitigating Algorithmic Bias in Hiring Systems

- Bias Detection
 - Analyze whether the model or data shows bias.
 - Use simple metrics (e.g., selection rate, accuracy by group).
- Impact Analysis
 - Discuss how bias affects individuals and organizations.
 - Provide real-world consequences (e.g., discrimination, legal risks).

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Case 3: Algorithmic Bias

Detecting and Mitigating Algorithmic Bias in Hiring Systems

- Proposed Solutions
 - Suggest at least 2 methods to reduce bias (e.g., data balancing, fairness constraints).
 - Explain how these solutions improve fairness.
- Ethical Reflection
 - Answer: “*Should this system be deployed? Why or why not?*”



Thank you for your attention!